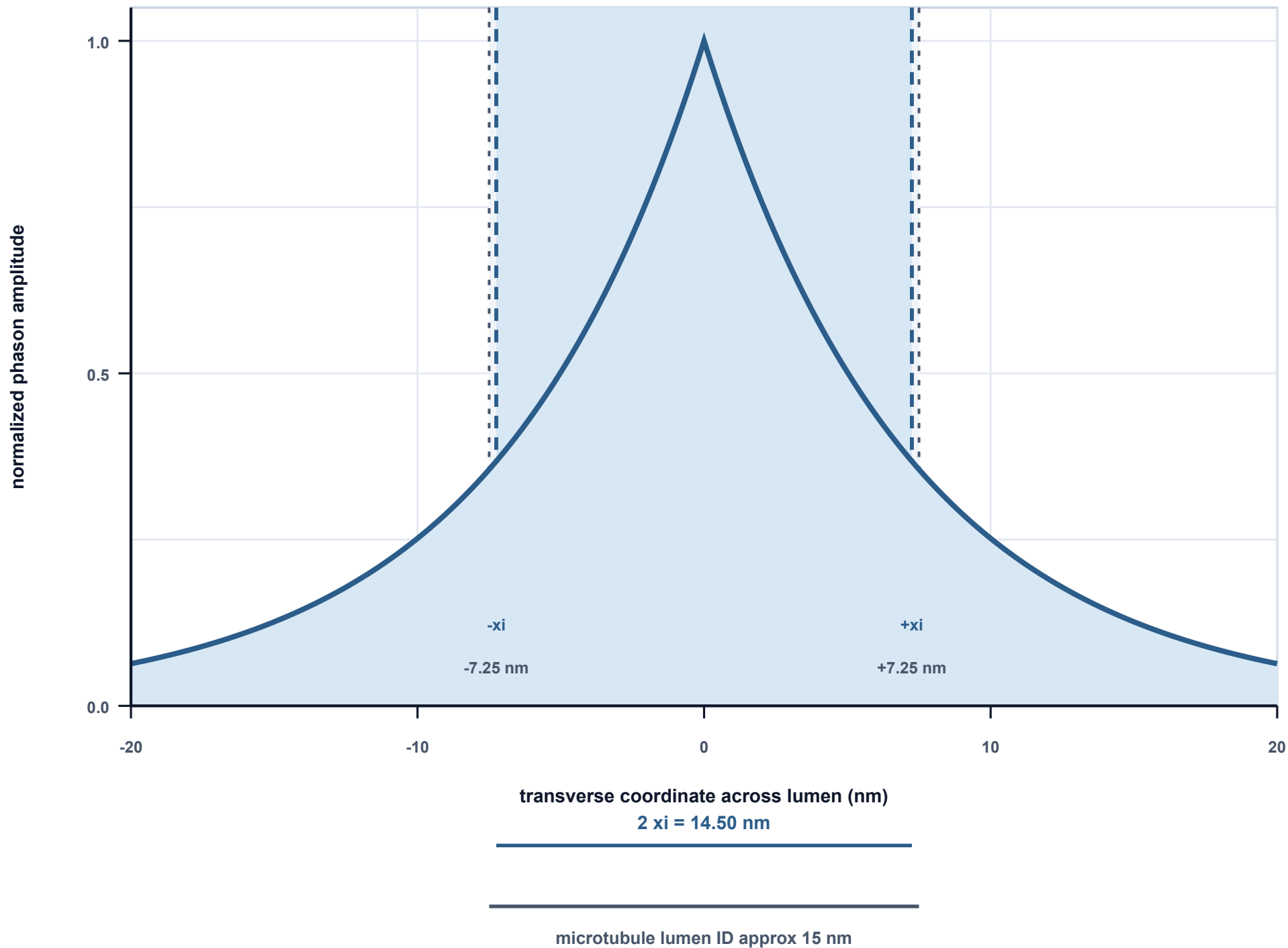


Figure V1.7.3

Phason healing length and lumen scale

Localized transverse profile: $\xi = a_0/\alpha$; 2ξ aligns with the microtubule lumen scale.



Source quantities

$a_0 = 0.05291772\text{ nm}$

$\alpha\text{ inverse} = 137.036$

$\xi = a_0 / \alpha = 7.25\text{ nm}$

$2\xi = 14.50\text{ nm}$

$K_{\text{perp}} / K_{\text{parallel}}$

equals ϕ power -18

$= 1.731\text{e-}04$

ξ : Tier 1 Bohr scaling

lumen match: Tier 3 correlation

Scale comparison



polaron diameter 14.50 nm



lumen diameter approx 15 nm

Interpretation: quantitative boundary-condition correlation only; not a first-principles derivation of the lumen geometry.